

Writing Small Numbers in Scientific Notation*Question Bank 3.09*

Show your work. Circle your final answer.

For questions 1 through 4, write each number in scientific notation.

1. 0.0005

2. 0.0000000049

3. 0.0316

4. 0.000000000529 meters (the radius of a hydrogen atom)

For questions 5 and 6, write each number in standard form.

5. 9×10^{-6}

6. 1.8×10^{-7}

Powers of ten.

7. Write each value as a single power of 10, then as a fraction or decimal.

a. 10^{-11} , as a fraction

b. $(10^4)^{-6} \times 10^{22}$

c. $10^{-5} \cdot 10^8$

8. Two measurements are 3.9×10^{-5} and 0.000000074.

a. Write 3.9×10^{-5} in standard form.

b. Write 0.000000074 in scientific notation.

c. Which of the two is greater? Explain how you know.

9. Complete each comparison.

a. How many times larger is 9.6 than 3.2?

b. How many times larger is 10^{-2} than 10^{-6} ?

c. How many times larger is 9.6×10^{-2} than 3.2×10^{-6} ?

10. Write each value in scientific notation.

a. 988 ten-thousandths

b. 0.00988

c. 988 millionths

11. Which is 0.00081 written in scientific notation?

- ☐ A. 81×10^{-5}
- ☐ B. 8.1×10^{-4}
- ☐ C. 8.1×10^{-3}
- ☐ D. 8.1×10^4

12. Select ALL of the expressions that are equal in value to 0.00052.

- ☐ A. 520×10^{-6}
- ☐ B. 0.52×10^{-3}
- ☐ C. 5.2×10^{-4}
- ☐ D. 5.2×10^{-3}
- ☐ E. 52×10^{-5}
- ☐ F. 5.2×10^4

13. Light travels about 3×10^8 meters per second. A snail travels about 3×10^{-4} meters per second.

a. How many times faster is light? Write the answer as a power of 10 and in standard form.

b. How many times slower is the snail than light?